

Optimization of double wishbone suspension system using GAs

Santhosh Kothapalli, NITW

30-11-2021

Abstract

Vibration isolation, stability and handling are the main design issues of any suspension. In this paper, a method to optimize a double wishbone suspension for minimizing bounce, roll, and comber variation and improve overall ride quality is presented. The suspension system is simulated along a bumpy track using point joint coordinates and is optimized using genetic algorithm. The simulation results for optimized and non-optimized systems are compared. A significant reduction in vertical acceleration and camber angle is observed.

Introduction

A suspension system includes the springs, dampers, linkages the connect the car body to the ground. The primary function of a suspension system is to keep the wheels in contact with the ground at all times and isolate the chassis from the road disturbances to provide good handling in rough terrains and improve ride comfort. The geometry and dynamic parameters like spring stiffness and damping ratio directly affect the vibration absorption capacity of the suspension.

Double wishbone and MacPherson strut are the most commonly used independent suspension systems. Double wishbone or A-arm suspension provides more control and stability compared to MacPherson strut since it boosts negative camber during bumps and while cornering. However, it has a more complex construction compared to MacPherson strut and finding the optimum geometry through trial and error can be difficult. Different types of vehicles are subject to various terrain conditions, for example – an ATV needs somewhat “loose” suspension than a passenger vehicle as it traverses in more uneven paths. The purpose of this study is to create a generic model suitable to obtain the suspension geometry for any type of four-wheeler.

Numerical model of Double wishbone suspension

The system is modelled using point-joint coordinates. Inputs in the form of elevation are given the left (u_l) and right (u_r) wheels. The positions, velocities and accelerations of all points are evaluated for each time step. A weighted function of camber variations, bounce of center of gravity and body roll is used as objective function for optimization.

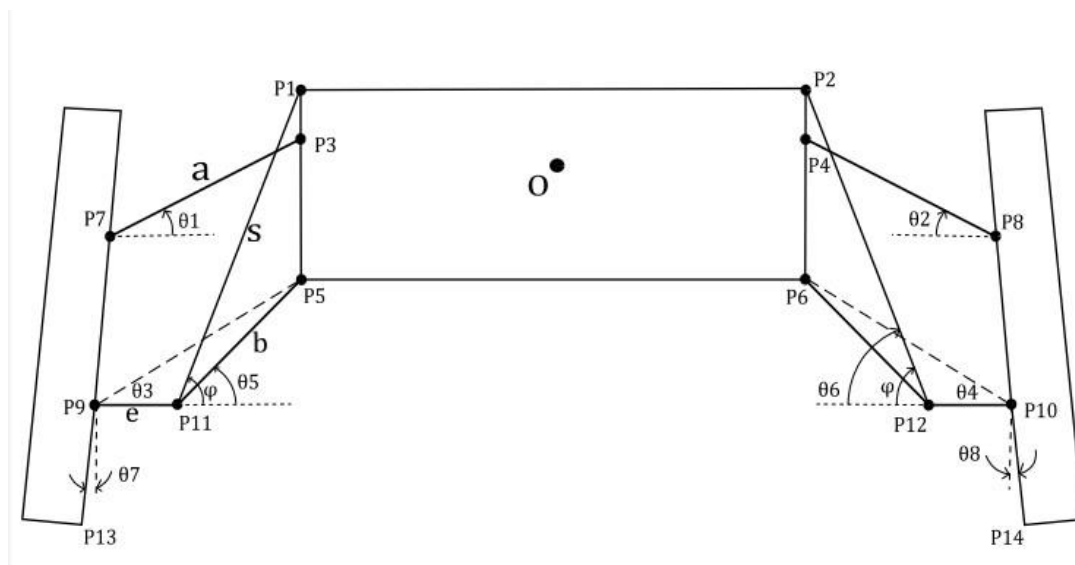


Figure 1: chassis and linkages

Kinematic equations:

the positions P(7-10) are calculated, keeping the chassis fixed, by solving the following non-linear equations using newton's method.

$$a \sin(\theta_1) + \left(R + \frac{d}{2}\right) \cos(\theta_3) - y_3 - h + u_{l(t)} = 0$$

$$b' \sin(\theta_3) + \left(R - \frac{d}{2}\right) \cos(\theta_7) - y_5 - h + u_{l(t)} = 0$$

$$x_3 - a \cos(\theta_1) - d \sin(\theta_7) - x_5 + b' \cos(\theta_3) = 0$$

Similar equations on right side give θ_2, θ_6 and θ_8 . Here, P9-P5-P11 acts as a rigid link. $b' = P9P5$

The new positions and velocities of unspring points are used in the governing equations to calculate the motion of spring points.

Governing equations:

$$\ddot{y}.m = k(y_{11} - y_{11(0)} - (y_1 - y_{1(0)})) + k(y_{12(0)} - y_{12(0)} - (y_2 - y_{2(0)})) + c((\dot{y}_{11} - \dot{y}_1) + (\dot{y}_{12} - \dot{y}_2))$$

$$\ddot{x}.m = k(x_{11} - x_{11(0)} - (x_1 - x_{1(0)})) + k(x_{12} - x_{12(0)} - (x_2 - x_{2(0)})) + c((\dot{x}_{11} - \dot{x}_1) - (\dot{x}_{12} - \dot{x}_2))$$

$$\ddot{\theta}.I = \left\{ k(y_{11} - y_{11(0)} - (y_1 - y_{1(0)})) - k(y_{12(0)} - y_{12(0)} - (y_2 - y_{2(0)})) + c((\dot{y}_{11} - \dot{y}_1) - (\dot{y}_{12} - \dot{y}_2)) \right\} \cdot f/2$$

where,

m = load on front axle (kg)

I = polar moment of inertia (kg-mm²)

$y_{i(0)}, x_{i(0)}$ indicates the positions at static load

k = spring stiffness

c = damping coefficient of spring

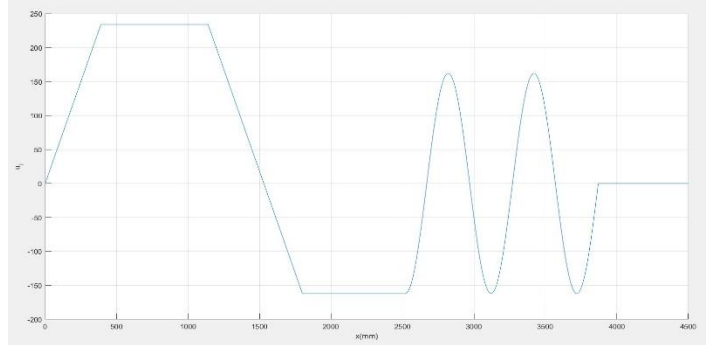


Figure 2: simulation track

Genetic Algorithms – A review

Genetic algorithm mimics the natural selection process and selects the best candidates based on nature inspired operations like selection, mutation and crossover. GAs have applications in many fields such as robotics, biomechanics, machine learning and economics.

A GA primarily has –

- A search domain represented in binary form
- A Fitness function

A sufficiently large population of candidates is initialized within the search domain. Selection probabilities proportional to the fitness value are assigned to each candidate and a parent set is selected to create next generation based of these probabilities.

In crossover operation, two parent points are randomly selected and their binary strings are crossed over at a random site based on the predefined cross-over probability (0.5-0.8).

In mutation, a random site in each binary string is flipped allowing the GA to explore new regions not covered by the parent generation. The mutation probability is usually very low (<0.08).

Using GAs to optimize suspension geometry

A search space is initialized with the following variables and limits

Parameter	Notation	range
Upper arm length	a	50 – 450
Lower arm length	b	50 – 450
Upper arm angle	α	-5 – 50
Lower arm angle	β	-5 – 50
Spring angle	ϕ	30 – 85
Uncompressed spring length	l	300-700
Spring compression under static load *	δ	40 – 200
Damping coefficient	c	100 – 400

* Spring stiffness coefficient can be used instead

These limits can be changed based on the vehicle design.

Sinusoidal, ramp, and horizontal line functions are used to simulate different tracks. Objective function is a combination of roll, bounce and camber variation.

A multi-objective algorithm may also be used, but the increased complexity and computation only results in very little improvement in the results.

Algorithm specifications:

No. of bits for each variable	50
Cross over	Single point, probability – 0.65
Mutation probability	0.05
Initial population	60
Max. number of generations	300

Results

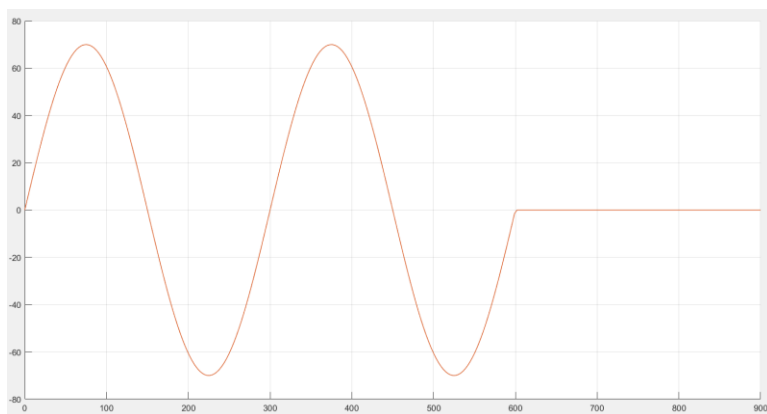


Figure: Test track

Baja SAE all-terrain vehicle –

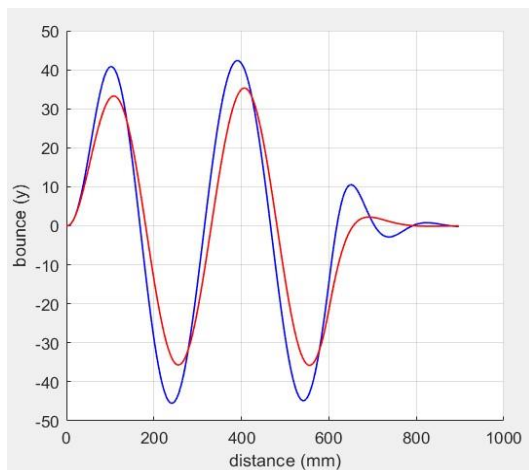
Height of Center of gravity = 700mm

Track Width = 1320mm

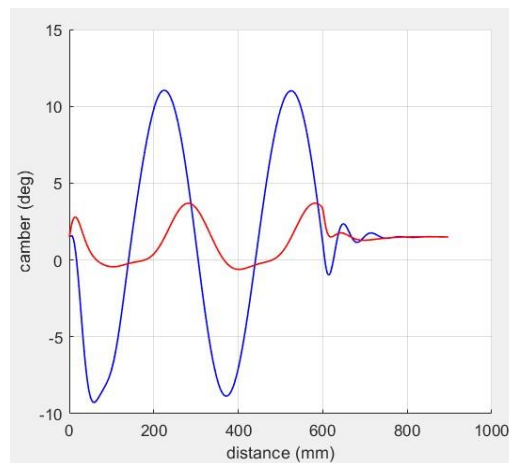
Wheel radius = 180mm

Load on front axle = 98kg

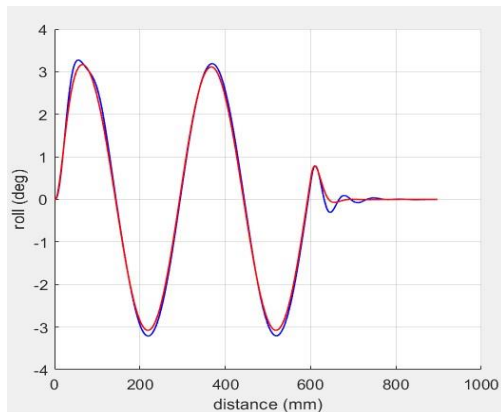
Design variable	units	Initial design	Optimized design
Upper wishbone length	mm	280	164.58
Lower wishbone length	mm	300	444.44
Upper arm angle	degrees	5	11.56
Lower arm angle	degrees	10	39.68
Spring elevation angle	degrees	70	84.4
Total spring length	mm	380	348.9
Spring compression under static load	mm	80	124.4
Damping coefficient	N-s/m	250	332.05



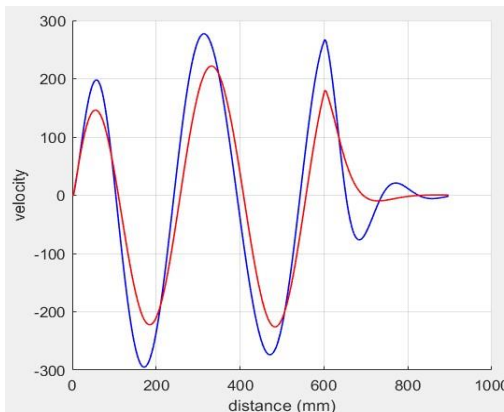
(a)



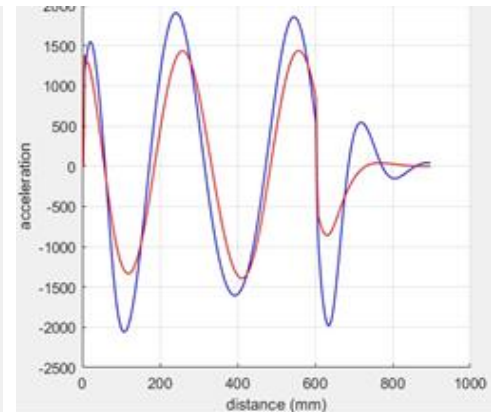
(b)



(c)



(d)



(e)

Figure 3: variation of (a) bounce, (b) camber, (c) roll angle, (d) vertical velocity, (e) vertical acceleration with distance along track for SAE-Baja ATV. Blue indicates unoptimized design and red indicated optimized design

Quadbike –

Height of Center of gravity = 510mm

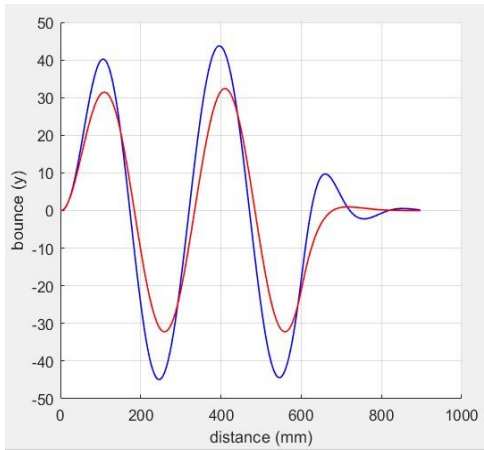
Track Width = 1200mm

Wheel radius = 250mm

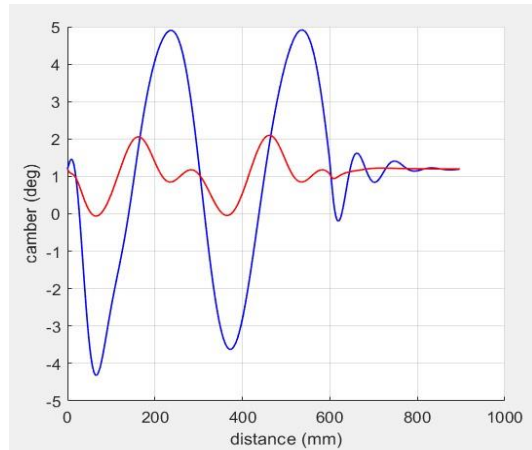
Load on front axle = 110kg

Design variable	Initial design	Optimized design
upper wishbone length	300	112.64
Lower wishbone length	300	327.01

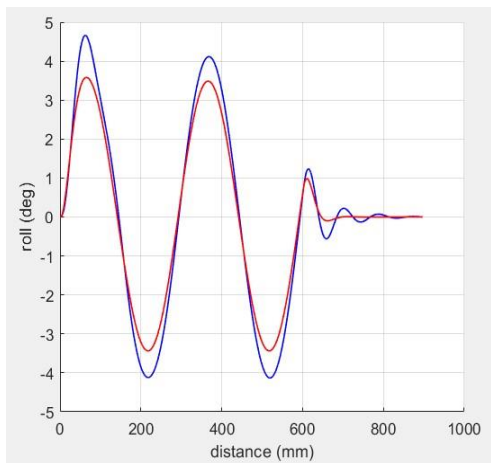
Upper arm angle	5	27.38
Lower arm angle	15	28.19
Spring elevation	65	83
Total spring length	420	370.37
Spring compression under static load	100	145.7
Damping coefficient	280	391.8



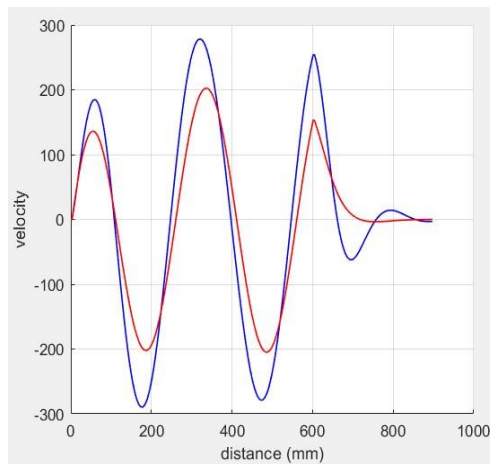
(a)



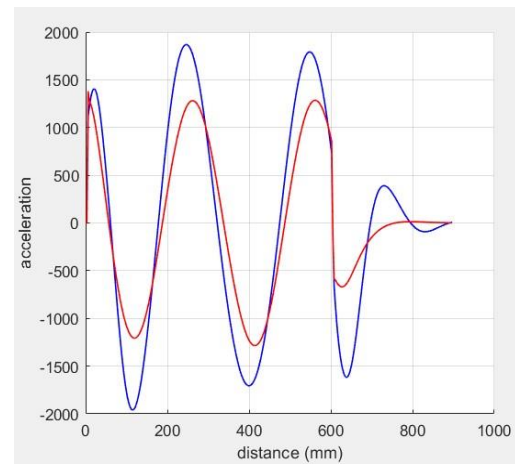
(b)



(c)



(d)



(e)

Figure 4: variation of (a) bounce, (b) camber, (c) roll angle, (d) vertical velocity, (e) vertical acceleration with distance along track for a quadbike. Blue indicates unoptimized design and red indicated optimized design